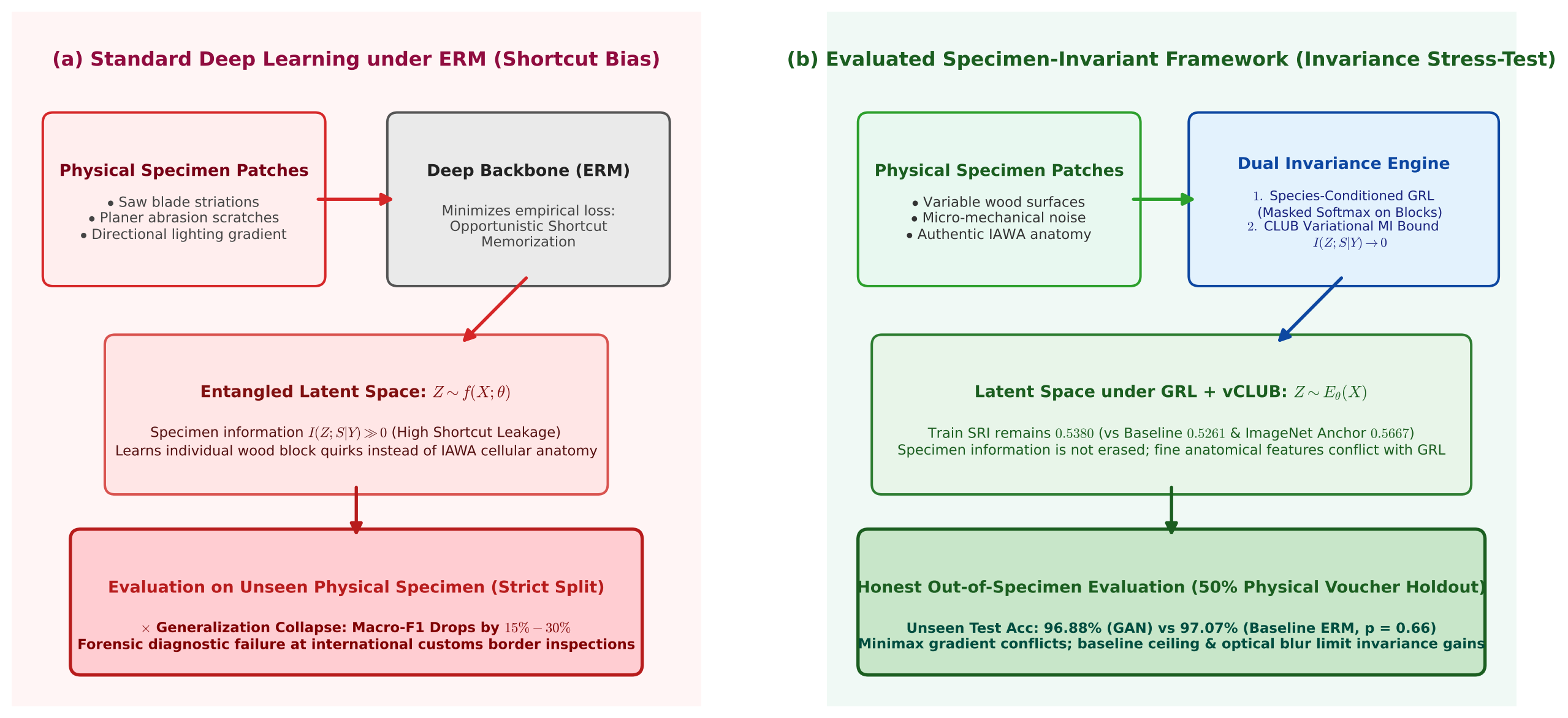

(a) Standard Deep Learning under ERM (Shortcut Bias)
Physical Specimen Patches
• Saw blade striations
• Planer abrasion scratches
• Directional lighting gradient
Deep Backbone (ERM)
Minimizes empirical loss:
Opportunistic Shortcut
Memorization
Entangled Latent Space: $Z \sim f(X;\theta)$
Specimen information $I(Z;S|Y) \gg 0$ (High Shortcut Leakage)
Learns individual wood block quirks instead of IAWA cellular anatomy
Evaluation on Unseen Physical Specimen (Strict Split)
× Generalization Collapse: Macro-F1 Drops by $15\% - 30\%$
Forensic diagnostic failure at international customs border inspections
(b) Evaluated Specimen-Invariant Framework (Invariance Stress-Test)
Physical Specimen Patches
• Variable wood surfaces
• Micro-mechanical noise
• Authentic IAWA anatomy
Dual Invariance Engine
1. Species-Conditioned GRL
(Masked Softmax on Blocks)
2. CLUB Variational MI Bound
$I(Z;S|Y) \to 0$
Latent Space under GRL + vCLUB: $Z \sim E_\theta(X)$
Train SRI remains 0.5380 (vs Baseline 0.5261 & ImageNet Anchor 0.5667)
Specimen information is not erased; fine anatomical features conflict with GRL
Honest Out-of-Specimen Evaluation (50% Physical Voucher Holdout)
Unseen Test Acc: 96.88% (GAN) vs 97.07% (Baseline ERM, p = 0.66)
Minimax gradient conflicts; baseline ceiling & optical blur limit invariance gains